BME402- ENGINEERING MECHANICS & STRENGTH OF MATERIAL

Course Outcomes: The student will be able to		Bloom's Taxonomy
CO1	Understand the force systems and application of force equilibrium to various two-dimensional problems.	**K3**
CO2	Understand the concept of stress and strain under different loading conditions.	**K3**
CO3	Determine the principal stresses and strains in structural members	**K3**
CO4	Understand and determine the stresses, slope, and deflection of the transversely loaded members	**K3**
CO5	Apply the concepts of stresses and strain in solving problems related to springs, buckling of columns and thin and thick cylinders.	**K3**

Unit I (10 Hours)

Introduction: Review of two-dimensional force systems, free body diagram, equilibrium of force systems, laws of friction, equilibrium analysis of simple systems involving friction.

Trusses: Introduction, simple truss and solution of a simple truss, methods of joints, methods of sections.

Beam: Introduction, shear force and bending moment, different equations of equilibrium, shear force and bending moment diagram for statically determined beams.

Unit II (8 Hours)

Centroid and moment of inertia: Centroid of plane, curve, area, volume and composite bodies, moment of inertia of plane area, parallel axis theorem, perpendicular axis theorem, concept and importance of principal moment of inertia.

Compound stress and strains: Introduction, normal stress and strain, shear stress and strain, stress on inclines sections, state of plane stress, principal stress and strain, maximum shear stress, Mohr's circle for plane stress, theories of failure, strain energy, impact loads and stresses, thermal Stresses, introduction to three-dimensional stresses.

Unit III (8 Hours)

Stresses in Beams: Pure Bending, normal stresses in beams, shear stresses in beams due to transverse and axial loads, composite beams.

Deflection of Beams: Differential equation of the elastic curve, cantilever and simply supported beams, Macaulay's method, area moment method, fixed and continuous beams.

Torsion: Torsion, combined bending and torsion of solid and hollow shafts, torsion of thin-walled tubes.

Unit IV (8 Hours)

Helical and Leaf Springs: Deflection of springs, helical springs under axial load and under axial twist (for circular cross sections), axial load and twisting moment acting simultaneously both for open and closed coiled springs, concept of laminated springs.

Columns and Struts: Buckling and stability, slenderness ratio, combined bending and direct stress, middle third and middle quarter rules, struts with different end conditions, Euler's theory for pin-ended columns, effect of end conditions on column buckling, Ranking Gordon formulae.

Unit V (8 Hours)

Thin cylinders & spheres: Introduction, difference between thin-walled and thick-walled pressure vessels, thin-walled spheres and cylinders, hoop and axial stresses and strain, and volumetric strain.

Thick cylinders: Radial, axial and circumferential stresses in thick cylinders subjected to internal or external pressures, compound cylinders, stresses in rotating shafts and cylinders, stresses due to interference fits.

Books:

1. Bansal, A Textbook of Engineering Mechanics, Laxmi Publications.
2. Bhavikatti and Rajashekarappa, "Engineering Mechanics", New Age International (P) Limited Publishers.
3. Meriam and Kraige, "Engineering Mechanics- Statics - Volume 1, 3ed., John Wiley & Sons.
4. Gupta S C, Strength of Materials, Pearson Education
5. Jindal, Strength of Materials, Pearson Education
6. Hibbeler, Mechanics of Materials, Pearson education.
7. Gere, Mechanics of Materials, Cengage learning.
8. Bere, Johnston, Mechanics of Materials, McGraw Hill.
9. Pytel, Mechanics of Materials, Cengage learning
10. Engineering Mechanics, Manoj, K. Harbola, Cengage learning
11. Strength of Material, J.K.Gupta, S.K.Gupta, Cengage learning

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